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Course Module Documentation

Controller Configuration

Attribute	Details
Area	Admin - Restricted to administrative users only
Authorization	AdminAuthorize attribute ensures proper authentication
Dependencies	- ICourseService: Core course business logic - ITrainerService: Trainer management - IUnitOfWork: Database operations - IMapper: AutoMapper for object mapping - IHttpContextAccessor: User context/session - IAuthorizationService: Permission validation - IOCRService: Document OCR processing - IHubContext<NotificationHub>: Real-time notifications - INotificationService: System notifications

Role-Based Access System

Feature	Details
User Type Detection	- Trainer: Restricted to own courses - Admin: Full system access - Auto-detection via <code>GetThisTrainerId_IfTrainer_else_0()</code>
Permission-Based Features	- Attendance requires "Course/Attendance" - UI adapts dynamically based on role - Selective data exposure for user types

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Core Course Management

Action	Details
Index	Main course listing, filtering, pagination, subscription tracking, optimized queries
AddEdit (GET/POST)	Course creation/modification, PDF validation, session-based return URL, file management
Delete	Removes course & subscriptions, cascade consideration, async operations

Member Subscription System

Feature	Details
MembersCourse	Shows members in course, attendance tracking, pagination, trainer info
SubscribedMembersInCourses	Subscription management, advanced filters, role-based data, optimized retrieval
Acceptance System	AJAX endpoint for accept/reject, JSON API, validation

Attendance Management

Feature	Details
Attendance Tracking	Real-time updates, boolean status, permission controlled, validation included

Reporting & Export System

Type	Details
Print	Course/member lists, certificates, professional formatting, filtered data
Excel Export	Role-appropriate columns, bilingual headers, dynamic column inclusion, timestamped files

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Feature	Details
PrintCertificate	Generates course completion certificates, integrates course/member data

Advanced Features

Feature	Details
Dynamic Trainer Loading	AJAX endpoint, department-based, bilingual names
Session Management	Return URL tracking, preserves filters & states
Search Implementation	Multi-field, case-insensitive, combined filters, optimized performance

Security & Validation

Feature	Details
File Validation	PDF only, 5MB limit, MIME type checks, temp storage, async handling
Model Validation	Full model state checks, CSRF protection, localized error messages
Permission Security	Role-based, dynamic permission checks, feature gating

API Design

Endpoint	Details
JSON Endpoints	Subscription accept/reject, attendance updates, trainer loading, permission checks
Response Format	<pre>{ "success": true/false, "message": "Descriptive message", "data": {} }</pre>

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CourseService Documentation

Operation	Details
GetAllAsync()	Retrieve all courses with related entities, eager loading, optimized queries
GetAllCoursesOfMemberAsync(memberId)	Retrieve member courses, active subscriptions only, includes department info
GetByIdAsync(id)	Fetch specific course with null safety
AddAsync(entity, file)	Create new course with optional file, atomic transaction, returns ID
UpdateAsync(entity, file)	Update course, file handling separated for controller flexibility
DeleteAsync(id)	Safely delete course after subscription checks, removes related files & subscriptions
DeleteSubscriptionAsync(id)	Delete individual member subscription with validation
SaveImageAsync(file)	Stores course attachments with unique naming, returns relative path
DeleteImageFile(path)	Removes physical attachment with safety checks

ViewModels

ViewModel	Purpose	Key Properties
AcceptanceVM	Handles subscription accept/reject	SubscriptionId, State, value, Notes
CertificateVM	Certificate data	Id, MemberNameAr/En, GenderId, CourseTitleAr/En, StartDate, EndDate
CourseVM	Course management	Id, DepartmentId, TrainerId, TitleAr/En, Dates, Location, Description, AttachmentPath, IsSubscribed
MembersCourseVM	Aggregated course & members	Course, Trainer, Members, Members_Paginated, Subscriptions
SubscribedMemberCourseVMTrainer	Trainer view of subscriptions	paginated, UserIsTrainer, trainerSelect, Trainer, TrainersList
SubscribedMemberCourseVM	Member subscription tracking	MemberFullNameAr/En, MemberIdNumber, CourseTitleAr/En, SubscriptionId, Acceptance, Attendance, TrainerFullNameAr/En

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DTOs

DTO	Purpose	Key Properties
AcceptanceDTO	Subscription accept/reject	SubscriptionId, State, value, Notes
CertificatedDTO	Certificate data	MemberNameAr/En, GenderId, CourseTitleAr/En, StartDate, EndDate
SubscribedMemberCourseDTO	Member subscription tracking	MemberFullNameAr/En, MemberId, SubscriptionId, Acceptance, Attendance, Course info, Trainer info

MembersCourse Module

Feature	Details
ViewModel	MembersCourseVM
Pagination	50,100,150 per page; previous/next buttons; maintains state via AJAX
Filters	Text search, nationality, date range, trainer
Attendance	AJAX updates via checkboxes, permission-controlled
Print/Export	Print views, Excel export, max 3999 records for print

Subscribed Members In Courses Module

Feature	Details
Filters & Search	Text search, date range, nationality dropdown, AJAX reload
Participant Table	Name, Nationality ID, Mobile, Email, Attendance/Approval; AJAX updates
Permissions	Delete, Accept, Rate, Edit member data
Pagination & Page Size	Previous/Next, 50/100/150 items, state preserved

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FougeraClub MVC Admin Summary

Module	Purpose	Key Notes
Account	User account management	ResetPasswordVM, SignatureVM, localized validation
Courses	Add/Edit courses	Multilingual titles, validation, file upload, AJAX trainer load
Validation & Permissions	Model & role-based validation	LocalizedRequired, NotInThePast, PermissionScanner